Critical Path Method (CPM) Analysis

200-Bed Hospital Construction (as-planned schedule, calculated by Planora)

The Critical Path Method finds the longest chain of dependent tasks. That chain sets the project duration: any delay to a task on it delays the whole project. Planora calculated this analysis automatically.

467 days	75	4 Jan 2027	18 Oct 2028
project duration (working days)	critical tasks (of 79)	project start	project finish

How the calculation works

1. **Forward pass.** Starting at the project start, each task's **Early Start (ES)** is the latest Early Finish of its predecessors; **Early Finish (EF)** = ES + duration (working time on the calendar).
2. **Backward pass.** Starting at the project finish, each task's **Late Finish (LF)** is the earliest Late Start of its successors; **Late Start (LS)** = LF − duration.
3. **Total float (slack)** = LS − ES. It is how long a task can slip without delaying the project.
4. **Critical path** = the tasks with zero total float.

Result by phase

WBS	Phase	Duration	ES	EF	LS	LF	Float
1	Pre-Construction	40 d	4 Jan 2027	26 Feb 2027	4 Jan 2027	26 Feb 2027	0 d
2	Site & Earthwork	25 d	1 Mar 2027	2 Apr 2027	1 Mar 2027	2 Apr 2027	0 d
3	Foundation	34 d	29 Mar 2027	13 May 2027	29 Mar 2027	13 May 2027	0 d
4	RCC Structure	83 d	11 May 2027	3 Sep 2027	11 May 2027	3 Sep 2027	0 d
5	Masonry	45 d	3 Sep 2027	5 Nov 2027	3 Sep 2027	5 Nov 2027	0 d
6	MEP First Fix	35.1 d	6 Oct 2027	24 Nov 2027	6 Oct 2027	24 Nov 2027	0 d
7	Plaster & Waterproofing	43 d	24 Nov 2027	24 Jan 2028	24 Nov 2027	24 Jan 2028	0 d
8	Doors Windows & Facade	35 d	12 Jan 2028	1 Mar 2028	12 Jan 2028	1 Mar 2028	0 d
9	Flooring & Finishes	72.1 d	1 Mar 2028	9 Jun 2028	1 Mar 2028	9 Jun 2028	0 d
10	MEP Second Fix	51 d	12 May 2028	24 Jul 2028	12 May 2028	24 Jul 2028	0 d
11	External Development	36 d	24 Jul 2028	12 Sep 2028	24 Jul 2028	12 Sep 2028	0 d
12	Testing Commissioning & Handover	33.1 d	1 Sep 2028	18 Oct 2028	1 Sep 2028	18 Oct 2028	0 d

The critical path

Every phase has zero float: the hospital plan is one continuous chain from **Site Survey & Soil Investigation** to **Project Handover**, with 75 critical tasks. Phases follow one another as:

Pre-Construction → Site & Earthwork → Foundation → RCC Structure → Masonry → MEP First Fix → Plaster & Waterproofing → Doors Windows & Facade → Flooring & Finishes → MEP Second Fix → External Development → Testing Commissioning & Handover → Project Handover

Tasks with float (not critical)

Task	Early finish	Late finish	Total float
Design Approval Milestone	19 Mar 2027	18 Oct 2028	412.4 d
Foundation Complete Milestone	6 Oct 2027	18 Oct 2028	270.3 d
Structure Topped Out Milestone	19 Apr 2028	18 Oct 2028	130.1 d
MEP Testing Complete Milestone	4 Jan 2027	18 Oct 2028	467.4 d

Only these milestones can slip; they are not linked to anything that follows them. Everything else is critical.

What this means for the project manager

- Because the whole plan is critical, **any delay** in any task moves the handover date. The plan has no buffer.
- To finish earlier, overlap phases (for example start MEP First Fix on lower floors while upper floors are still being built) or add crews to the longest phases: **RCC Structure** and **Flooring & Finishes**.
- Track the critical tasks weekly; Planora shows them in red when the Critical path switch is on.

Source. Hospital_Construction_Advanced.xml from the construction test pack, opened in Planora 1.1.0. Dates are the **as-planned** schedule (progress recorded in the file cleared), calculated by Planora on the file's calendar (Mon–Fri, 8:00–17:00). Two links in the file could not be imported (one to the project summary row, one to a missing task); Planora reports them as import notes. Amounts are in the file's currency units, shown as ₹.